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RESEARCH ARTICLE

Sign Language Recognition Using Graph and General Deep Neural Network Based on Large Scale Dataset

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ABSTRACT Sign Language Recognition (SLR) represents a revolutionary technology aiming to establish communication between hearing impaired and non-hearing impaired communities, surpassing traditional interpreter-based approaches. Existing efforts in automatic sign recognition predominantly rely on hand skeleton joint information, steering clear of image pixels to address challenges like partial occlusion and redundant backgrounds. Many researchers have been working to develop automatic sign recognition using hand skeleton joint information instead of image pixels to overcome partial occlusion and redundant background problems. However, body motion and facial expression play an essential role in increasing the inner gesture variance in expressing sign language emotion besides hand information for large-scale sign word datasets. Recently, some researchers have been working to develop multi-gesture-based SLR recognition systems, but their performance accuracy and efficiency are unsatisfactory for real-time deployment. Addressing these limitations, we propose a novel approach, a two-stream multistage graph convolution with attention and residual connection (GCAR) designed to extract spatial-temporal contextual information. The multistage GCAR system, incorporating a channel attention module, dynamically enhances attention levels, particularly for non-connected skeleton points during specific events within spatial-temporal features. The methodology involves capturing joint skeleton points and motion, offering a comprehensive understanding of a person's entire body movement during sign language gestures and feeding this information into two streams. In the first stream, joint key features undergo processing through sep-TCN, graph convolution, deep learning layer, and a channel attention module across multiple stages, generating intricate spatial-temporal features in sign language gestures. Simultaneously, the joint motion is processed in the second stream, mirroring the steps of the first branch. The fusion of these two features yields the final feature vector, which is then fed into the classification module. The model excels in capturing discriminative structural displacements and short-range dependencies by leveraging unified joint features projected onto a high-dimensional space. Owing to the effectiveness of these features, the proposed method achieved significant accuracies: 90.31%, 94.10%, 99.75%, and 34.41%, for the WLASL, PSL, MSL, and ASLLVD large-scale datasets, respectively, with 0.69 million parameters. The high-performance accuracy, coupled with stable computational complexity, demonstrates the superiority of the proposed model. This innovative approach is anticipated to redefine the landscape of sign language recognition, setting a new standard in the field.

INDEX TERMS Sign language recognition (SLR), American sign language (ASL), large scale dataset, hand pose, graph convolutional network (GCN), graph convolutional with attention and residual connection (GCAR), deep learning network.

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I. INTRODUCTION

Primary communication tool for the hearing impaired community is sign language. This is different from the

common language and has a different linguistic structure. The hard of hearing and non verbal community cannot express their thoughts, ideas, and requirements with general languages such as English, Hindi, Urdu, and Bangla because they cannot talk with their disability [1], [2], [3], [4], [5]. In this situation, they used a spatial kind of sign, posture, or symbol, including facial, body, head, hands and finger posture information, known as sign language [6], [7], [8], [9]. The number of hearing impaired; people at work has increased daily. Based on the information, 5% of the world's population, specifically 466 million people around the world, belong to this community where 432 million are adults and 34 million are children [10]. So, the hearing impaired community need to learn this sign language to express their requirements and basic needs to others and establish a communication link with their hearing impaired and non-hearing impaired community for basic needs such as education, medical services, and employment [11]. The main problem is that it is difficult to learn symbolic sign language because of its complexity. Moreover, though the hearing impaired community tries hard to learn this language, the non-hearing impaired community is not interested in learning this language to establish a communication link with that community. One of the main causes of the complexity of sign language is that it is not uniform for the whole world, but it is different from country to country and culture to culture. Even in one country, there are severe types of sign language. For example, India has Indian sign language, Assam sign language, etc. America and Britain both societies use the English language, but there are different sign languages, namely American Sign Language (ASL) and British Sign Language (BSL). Furthermore, there is a lack of awareness about sign language among many hearing impaired and mute individuals, despite living in a modern society where the majority of people use smartphones with internet access. An expert or human sign language translator may be the only effective solution to the numerous complexities of learning this language because it is essential to use. The problem is that it is still challenging to find a human sign language translator, and it needs to pay for high-cost events if it needs a short time.

In this situation, the automatic sign language translator method is inevitable to establish communication for the hearing impaired and dumb community. In recent years, many researchers have developed a system to create a new way of communication by transferring sign gestures into corresponding meaning text, which can help the hearing impaired community establish fluent communication with other hearing impaired and non-hearing impaired communities. Developing an effective sign language recognition system is challenging as it faces the hearing impaired community to learn it because it consists of a series of actions that include quick motion and similar characteristics. Based on the sign meaning of this language, categorization into two types is undertaken: static or dynamic signs. This classification aligns with established frameworks found in prior research published in reputable journals. A single

image containing the full meaning of the sign is known as a static sign, and if it needs more than one image or sequence of images, it is known as a dynamic sign. The main problem of static sign language recognition is the large variation of the vocabulary set and the large variation in the hand actions. Moreover, some significant variations from the signers can cause a misunderstanding about the actual meaning of the signs. In this situation, dynamic sign language recognition is more effective and efficient in overcoming the related challenges. Effective sign language information or potent data collection is inevitable to develop a good sign language recognition system. Till now, researchers have used two types of data collection methods to develop the SLR system, namely the sensor-based [12] system and the vision-based [13] system. Sensor-based systems can collect sign information from signers effectively, but cost and portability are common problems of this approach. Vision-based sign language recognition systems lead to challenges in sensor-based systems, considering the scalability, flexibility, and low-cost approach, which make them popular among researchers. Most of the computer vision researchers have been working to develop a sign language recognition (SLR) system with two varieties: pixel-based image [3], [4], [14] and skeleton point of the image [14], [15]. In the image-based system, they took RGB images as the input and applied different machine learning and deep learning on it either on the handcrafted feature or directly on pixel values. However, they still face problems in achieving high-performance accuracy and efficiency because of the various complexities, including background complexity, redundant background, partial occlusion, lighting conditions, environmental complexity, orientation, etc. To overcome this challenge, the researcher used the skeleton point of the image instead of the pixel point, which is mainly a sequence of the 2D or 3D coordinates of the hand gesture. The hand joints are mainly recorded from various kinds of spatial cameras, but recently used various kinds of deep learning-based pose detection software such as MediaPipe, Alpha Pose, OpenPose, etc. After extracting the skeleton point, many researchers employed hand-crafted feature extraction techniques, including trajectory, HOG [16], SIFT [17] and then applied machine learning and deep learning algorithms. Shin et al. applied the geometrical method to extract distance and angular features from the 21 hand key points, which are extracted using the MediaPipe system for the ASL dataset [18]. The main limitations of the handcrafted feature-based system are inefficient features and a lack of generalization. Recently, many researchers employed deep learning algorithms to classify hand gestures using raw skeleton information [19], [20], [21], [22]. These existing systems only consider the spatial information of the frame and do not consider the motion and temporal features, and in most cases, they fail to extract intricate relationships among the joints. To overcome the problems, researchers used Musa et al. to employ a sophisticated graph convolutional network (GCN) and attention-based spatial, temporal network across the skeleton point to extract the

intricate relationship of the hand skeleton [14]. Although these methods can show their excellence in sign language recognition in some cases, in most cases, they may fail to achieve satisfactory performance. The main drawback of the system is that they used only hand information, which may not express the full meaning of the sign because of the inefficient expression of information. To interpret a sign and bring the actual meaning of the sign, it is important to collect whole body information, including facial, body and hand pose. In summary, only hand pose information may be insufficient to correctly convey the sign information because of the lack of expression and emotional information, which can be conveyed in the whole body skeleton by carrying useful information about the signer's expression. Few researchers have considered the body and face besides hand gestures for sign language recognition. Among them, Amorim et al. developed 67 skeleton key points based on 20 classes of the ASLLVD dataset, which include word level sign language word recognition using the enhancement of [22] by adding graphic layout [23]. Solis et al. collected 30 Mexican sign language (MSL) word skeletons considering body, face and hand information using a spatial camera, then applied RNN and LSTM and achieved good accuracy [15].

In the same way, other researchers develop 67 whole body key points based on deep learning methods such as RNN [24], LSTM [25]. The main drawback of the mentioned key point-based sign language recognition is that, in most cases, they used small and self-development datasets to evaluate their model and achieve good accuracy, and these systems may fail to achieve high-performance accuracy with large-scale datasets. In addition, they used the static sign language recognition method, which led to the system making generalizations, and it will not work for the dynamic sign. To overcome the lack of an effective dataset, Li et al. make a large-scale ASL world dataset by including 2000 classes or sign word [17]. Then, they applied a temporal graph convolutional network (TGCN) for the feature extraction and classification and achieved 87.60% at top-10 accuracy for the WLASL 100 classes. Recently, Tunga et al. [26] employed Graph convolution with Bert, and they achieved 88.67% accuracy for the top-10 accuracy for the WLASL 100 classes dataset. In addition, Matyas [27] employed a spotter model, and they only show a few class label configurations of the WLASL dataset. Another large-scale dataset, ASSLLVD [23], is used by many researchers [23], but their system is still facing problems in achieving high-performance accuracy [23], [28]. The main drawback of the systems is lower performance accuracy and may fail to achieve the system's generalised properties due to a lack of effective features. In addition, they did not consider the role of non-connected skeleton points in fall events during the spatiotemporal feature extraction. However, the positional relationship of some non-real connected points is very helpful for partially identifying events. Moreover, most of the research work for whole-body skeleton-based SLR is related

to the static sign. There is no motion information based on the whole body skeleton included, and a few researchers have been working to develop dynamic SL recognition by including motion and GCN. A two-stream graph convolution with attention and residual connection (GCAR) is proposed in this study to address current challenges and advance medical technology systems. The work contributes significantly to the Sign Language Recognition Systems (SLRS) field, particularly in overcoming challenges related to generalization, accuracy, and efficiency. The contributions of this study are outlined as follows:

- **Multistage GCAR Based Two Stream Module:** A novel two-stream model is introduced, incorporating a multistage Graph Convolution with Attention and Residual connection (GCAR). This architecture adeptly captures spatial-temporal contextual information, overcoming challenges in generalizing models across diverse datasets. Incorporating a Channel Attention Module dynamically enhances attention levels, particularly for non-connected skeleton points during specific events in spatial-temporal features, thus empowering the model to focus on pertinent information and enhance discriminative capabilities.
- **Incorporation of Separable TCN and Channel Attention:** In the first stream, a synergistic combination of a Separable Temporal Convolutional Network (Sept-TCN), a GCN module, and a potent channel attention module is employed. This integration facilitates the extraction of discriminative features, emphasizing structural displacements and short-range dependencies. Similarly, features were extracted from the second stream, which took joint motion information as the input dataset. The final powerful feature was obtained by fusing the two features and fed into the classification module.
- **Extensive Experimentation and Generalizability:** The proposed method significantly advances the field through a robust feature fusion approach, a pivotal factor in boosting performance accuracy. Extensive experimentation encompassing diverse datasets, including a large-scale sign language dataset with varying configurations, underscores the model's robustness. "Four benchmark datasets, including two large-scale ASL dynamic datasets, were utilized, achieving accuracy rates of 90.31%, 34.41%, 94.10%, and 99.75% for WLASL, ALLVD, PSL, and MSL Dataset, respectively. Remarkably, proposed approach surpasses existing models in terms of accuracy, establishing its superiority and versatility as a more generalizable solution in sign language recognition. This positions the model as an efficacious and groundbreaking standard in the field.

The presented work is organized as follows: Section II summarizes the existing research work and problems related to the presented work, Section III describes the benchmark and proposed Korean sign language datasets, and Section IV describes the architecture of the proposed system—Section V demonstrated the evaluation performed

with state of the art comparison. In Section VI, draw the conclusion and future work.

II. RELATED WORK

Numerous investigations have been conducted in the sign language recognition domain. Recently, some research has published a new variation of the sign structure to establish the sign structure in perceiving subunits analogous to phonology in spoken language structure in perceiving subunits analogous to phonology in spoken language [29]. The Stokoe paradigm can be used to define sign language (SL) components, which rely on factors such as shape, alignment, and motion [30]. The palm orientation with the layout of the fingers can be expressed by the hand shape, which mainly carries a specific meaning and is captured with an image frame. This hand shape can be static and dynamic. The static posture can carry the full meaning of the sign with the single frame that is also a continual signing [29]. The sequential frames need to express the full meaning of the dynamic gesture, which is also known as the movement-hold model. While the correlation between the motion and stances of the hands has been approximated in two-handed signs, the aforementioned approaches are satisfactory for one-handed signs. To record the hand gesture-based sign language gesture, researchers mainly used vision-based sensors and sensor-based systems for SLR. Researchers focus on the vision-based system due to various advantages, including ease of use, portability, low cost, etc. There are many researchers who have been working to develop SLR systems using numerous statistical and mathematical models besides the various machine learning algorithms [3], [4], [14], [29], [30], [31]. Raghuveera et al. applied the HOG feature extraction technique to extract features from the RGB image for the ISL dataset and achieved 78.85% accuracy using the SVM algorithm [32]. Solis et al. extracted the Jacobi-Fourier Moments (JFMs) feature from the RGB image and achieved 95.00% accuracy using ANN for the MSL dataset [33]. The machine learning algorithm is good for small data sets with minimum time and computational complexity. After that, researchers thought about the large-scale dataset where the machine learning algorithm was not able to prove excellent for this, and researchers focused on the deep learning-based model. Miah et al. applied a CNN model on the segmented ASL dataset and achieved good performance accuracy compared to the existing system [3]. Neverova et al. proposed a spatial feature extraction mechanism using the ModDrop framework where they focus their multimodality, including RGB image, audio, etc. [34]. The main limitation of their work is that they integrated an audio data set, which has not yet been validated in the SLR domain. Put et al. proposed a multi-stage technique in the initial phase, specifically a sequence modelling technique known as Connectionist Temporal Classification (CTC). In the second stage, they employed a feature learning model: a 3D Convolutional Residual Network (3D-ResNet). They jointly trained the Long Short-Term Memory (LSTM) model

with CTC using a soft Dynamic Time Warping (soft-DTW) alignment constraint. They utilized the CSL and RWTHPHOENIX-Weather datasets to evaluate their model, achieving word error rates (WER) of 36.7% and 32.7%, respectively [35].

Koller et al. integrated two distinct features, incorporating Convolutional Neural Network (CNN) and Hidden Markov Model (HMM) sequence features. Subsequently, they applied a CNN-HMM-based hybrid model for classification. The evaluation of their model involved three benchmark sign language datasets, and they attained high-performance accuracy, reducing the WER by 20% [36]. Huang et al. introduced a Sign Language Recognition (SLR) model that involves extracting spatial-temporal features and selecting highlight features using 3D-CNN and an attention model [37]. Their model was evaluated on two benchmark datasets, ChaLearn-14 and CSL, reporting an accuracy of 95.30% for the ChaLearn dataset. Pigou et al. proposed a pooling-based temporal feature extraction method from a sign language video dataset [38]. Sincan et al. developed a hybrid method for effective feature extraction by combining LSTM, CNN, and a feature pooling approach [39]. They utilized VGG-16 as a pre-trained model integrated with a CNN-based parallel method for the Depth and RGB-based video dataset, achieving 93.15% accuracy with an Italian sign language dataset. Huan et al. introduced a temporal segmentation technique, Hierarchical Attention Network with Latent Space (LS-HAN), to enhance continuous sign language recognition. The goal was to increase accuracy and reduce computational complexity by eliminating preprocessing techniques [40], [41]. They designed a two-stream CNN for training and prediction using LS and HAN video features, achieving better accuracy than previous studies. However, their work has a limitation as it only considers pure visual features, which may not be suitable for capturing body movement and sign language nuances. The mentioned image-based SLR system has many merits and advantages. Still, these methods have been facing a lot of problems in achieving high-performance accuracy and efficiency due to redundant background, hand occlusion, managing illumination variations, and addressing hand orientation. To overcome the problem, researchers used skeleton points from the image instead of the direct use of pixels of the image for hand gesture recognition. Recently, researchers used various spatial kinds of 3D cameras for collecting skeleton points from the sign language image data; researchers have developed numerous software to extract the skeleton points, such as Alphaspose, OpenPose, MediaPipe, MMpose, etc. Shin et al. applied the geometrical method to extract distance and angular features from the 21 hand key points, which are extracted using the MediaPipe system for ASL dataset [18]. The main limitations of the handcrafted feature-based system are inefficient features and a lack of generalization. To overcome the limitation, many researchers employed end-to-end deep learning algorithms to classify hand gestures using raw skeleton information [19], [20], [21]. These existing systems only consider the

spatial information of the frame and do not consider the motion and temporal features. In most cases, they fail to extract intricate relationships among the joints. Musa et al. employ a sophisticated graph convolutional network (GCN) and attention-based spatial-temporal network across the skeleton point to extract the intricate relationship of the hand skeleton [14]. Yan et al. employed the Alpha pose to extract 21 hand skeleton joints and then applied Spatio-temporal GCN to recognize the hand gesture recognition [22]. These methods can show their excellence in sign language recognition in some cases, but they may fail to achieve satisfactory performance in large-scale dataset cases with a high value in the inner gesture similarity. The main drawback of the system is that they used only hand information, which may fail to make high variance patterns among the inter gestures and not express the full meaning of the sign because of the inefficient expression of information. It is important to collect whole-body information, including facial, body and hand poses, to increase inter-gesture variance, and it can help bring the sign's actual meaning. Few researchers have considered body and face besides hand gestures to reduce inter-gesture similarity for large-scale dataset-based sign language recognition. Among them, Amorim et al. developed 67 skeleton key points based on 20 classes of the ASLLVD dataset, which include word level sign language word recognition using the enhancement of [22] by adding graphic layout [23]. Solis et al. collected 30 Mexican sign language (MSL) word skeletons considering body, face and hand information using a spatial camera, then applied RNN and LSTM and achieved good accuracy [15]. Xia et al. made a dataset by considering 67 whole body key points and achieved satisfactory performance using RNN with their self-development dataset [24]. Peraz et al. applied LSTM to develop a gesture recognition system with the 67 whole-body skeleton points and achieved high performance with their dataset [25]. The main drawback of the mentioned whole key point-based sign language recognition is that in most of the cases, they used small and self-development datasets to evaluate their model and achieve good accuracy, and these systems may fail to achieve high-performance accuracy with large-scale datasets. In addition, they used the static sign language recognition method, which led to the system for making generalizations, and it will not work for the dynamic sign. To overcome the lack of an effective dataset, Li et al. make a large-scale ASL world dataset by including 2000 classes or sign word [17]. Then, they applied a temporal graph convolutional network (TGCN) for feature extraction and classification and achieved 62.63% at top-10 accuracy. In this study, a multistage graph convolution with attention and residual connection (GCAR) model is proposed to address issues related to enhancing the accuracy and effectiveness of sign language recognition systems.

III. DATASETS

In this study, different benchmark sign language datasets, namely WLASL, MSL, ASLLVD, and PSL, are utilized to

TABLE 1. Evaluated dataset description in the study.

Dataset Names	Lang.	Signs	Sub.	Total videos	Videos Per Sign	Joint Per Frame
WLASL	ASL	2000	119	21089	10.5	67
MSL	MSL	30	20	3000	20	67
ASLLVD	ASL	2745	n/a	9748	3/4	67
PSL	PSL	19	n/a	2700 (img)	55 (img)	67

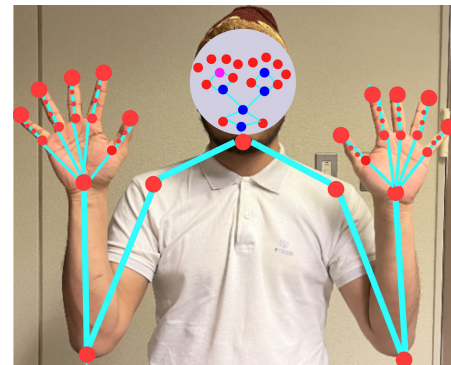


FIGURE 1. Pose and graph construction.

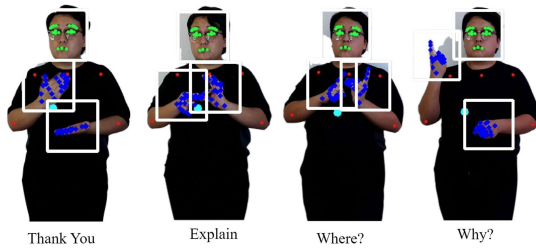
evaluate the proposed model and the dataset description are described in the following subsections.

A. WLASL DATASET

WLASL stands out as one of the largest scale datasets for ASL that consists of 68,129 videos featuring 20,863 ASL glosses collected from 20 diverse websites [17]. Each video captures a signer performing only one sign, sometimes recorded only one time and sometimes repeated, and most of the video is recorded from the frontal view but considered a diverse, complex background. Following the dataset compilation, videos associated with gloss annotations containing more than two English words are excluded, ensuring that the dataset comprises solely single-word annotations. The removal of gloss, lacking seven videos, was implemented to ensure an adequate sample for the division of training and testing sets. Given that most websites incorporate commonly used words, the limited number of video samples for certain glosses may indicate infrequent usage. Thus, the removal of glosses with minimal video samples has a negligible impact on the practical utility of the dataset. After completing the pre-selection and processing, This dataset is composed of 34404 videos for 3126 glosses for further annotation, which are organized by sorting glosses in descending order based on sample numbers. Experiments are conducted on datasets with varying vocabulary sizes to offer insights into the challenges of word-level sign recognition and the scalability of sign recognition methods. Specifically, the top-K glosses are selected with K values of 100, 300, 1000, 2000, resulting in four subsets named WLASL100, WLASL300, WLASL1000, and WLASL2000, respectively.

TABLE 2. The description of the 67 whole body key points.

Sr No	Hand Pose No	L. Hand Pose Name	Sr No	Hand Pose No	R. Hand Pose Name	Sr No	Hand Pose No	Right Hand Pose Name
1	0	Wrist	20	19	Ring MCP	36	9	Middle CMC
2	1	Thumb CMC	21	20	Ring IP	37	10	Middle MCP
3	2	Thumb MCP			Body Pose Name	38	11	Middle IP
4	3	Thumb IP	22	0	Nose	39	12	Middle TIP
5	4	Thumb TIP	23	11	Left shoulder	40	13	Ring CMC
6	5	Index CMC	24	13	Left elbow	41	14	Ring MCP
7	6	Index MCP	25	12	Right shoulder	42	15	Ring IP
8	7	Index IP	26	14	Left elbow	43	16	Ring TIP
9	8	Index TIP			Hand Pose No	44	17	Right heel
10	9	Middle CMC	27	0	Wrist	45	18	Ring CMC
11	10	Middle MCP	28	1	Thumb CMC	46	19	Ring MCP
12	11	Middle IP	29	2	Thumb MCP	47	20	Ring IP
13	12	Middle TIP	30	3	Thumb IP			Facial Landmark
14	13	Ring CMC	31	4	Thumb TIP	48-51	4	Right eyebrow
15	14	Ring MCP	32	5	Index CMC	52-55	4	Left eyebrow
16	15	Ring IP	33	6	Index MCP	56-59	4	Right eyebrow
17	16	Ring TIP	34	7	Index IP	60-63	4	Left eyebrow
18	17	Right heel	35	8	Index TIP	64-67	4	Mouth
19	18	Ring CMC						

**FIGURE 2.** Sample images extracted from the Mexican Sign Language (MSL) dataset.

B. MSL DATASET

This dataset was also collected by including two hands, a body, and a face in 67 key points. They used a special camera, namely an OAK-D camera, to record this dataset by including the MediaPipe library. This dataset was collected from 30 different MSL signs with 25 repetitions for each sign. They recorded 3000 samples for the dataset in total, specifically 20 videos for each sign and extracted 20 frames from each video [15]. Figure 2 demonstrated the sample example of the MSL Dataset.

C. ASLLVD DATASET

Another large sign language word dataset name is ASLLVD, which is an abbreviation of the American Sign Language Large-Scale Video Dataset (ASLLVD dataset). This dataset contained 2745 ASL sign words which were recorded from multiple synchronizations with different angles. In the dataset, there are 7798 videos for training and testing, and there are 1950 videos. Each video has a start and end frame, the class labels, which mainly form gloss labels. It consists of several synchronized videos showing signing from different

**FIGURE 3.** Sample images from American Sign Language Lexicon Video Dataset (ASLLVD).

angles, with hand shapes labelled with morphological and articulatory classifications. Each morpheme included different annotations for the compound signs. In order to enhance the recognition of sign language through computer vision, the dataset incorporates unique numerical ID labels for different sign variations, uncompressed raw video sequences, and sequences for camera calibration, as discussed in [23]. Figure 3 demonstrated the sample example of the ASLLVD Dataset.

D. PAKISTANI SIGN LANGUAGE DATASET

This is another sign language dataset that also includes body and face information besides the hand. This dataset is mainly developed for the Pakistani sign language dataset, which is captured from the whole-body skeleton point using MediaPipe and OpenPose system and web camera and saved as a JSON file. There are 67 skeleton key points in these datasets they extracted from each frame. In the dataset, they collected x, y, and z coordinate values for the 67 key points, where 21 from the right hand and 21 from the left hand and body with face extracted 25 skeleton key points. This dataset consists of alphabets and words, with 12 word signs and 37 alphabet signs. It is recorded from nine people for both the alphabet and words. After recording the dataset, they normalized it with hand and body positions. Figure 1 demonstrates the 67 key points in the whole body. **Copyright ©2020 Saad Butt and the dataset is licensed under the GPL3.0 license.** The dataset was obtained from the Kaggle link available at <https://www.kaggle.com/datasets/saadbutt321>.

IV. PROPOSED METHODOLOGY

Let the sequence of the input pose is $X_{(1:N)} = [x_1, x_2, x_3, \dots, x_N]$ in N sequential frames. Here, concatenated skeleton key points are represented by $x_i \in R^K$ in dimension K . In this study, we propose a two-stream model, utilizing a multistage graph convolution with attention and residual connection (GCAR) to extract spatial-temporal contextual information. The model leverages a channel attention module to dynamically enhance attention levels, especially for non-connected skeleton points during specific events in spatial-temporal features. The process involves capturing joint skeleton points and joint motion to identify a person's entire body movement during sign language gestures, feeding information into two model streams.

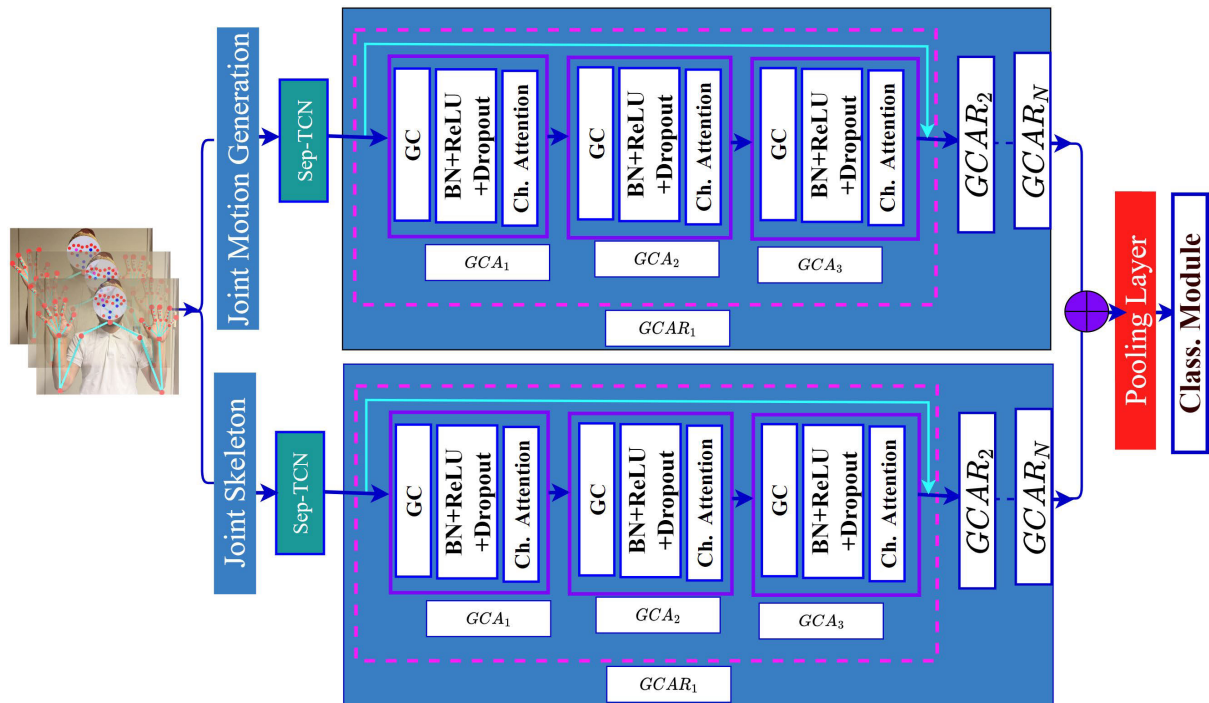


FIGURE 4. Working flow architecture.

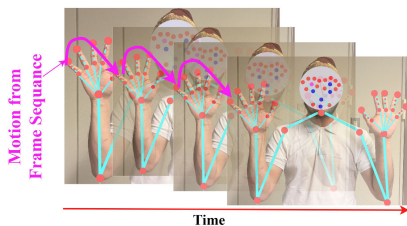


FIGURE 5. Motion calculation demonstration.

In the first stream, joint key points undergo processing through Sept-TCN, followed by a Sep-TCN and GCN module. The GC module's output then enters a deep learning module with multiple layers, and a channel attention module selects channel-wise effective features for the GCA module. Subsequently, three GCA modules are sequentially employed, with the third module's output concatenated with the residual connection to produce features for the first stage of the GCAR module. A total of 22 stages of GCAR in both streams generate feature vectors from joint key points.

In the second stream, joint motion serves as input, and features are generated similarly to the first branch. The combination of these two feature vectors creates the final feature vector fed into the classification module. The model captures discriminative structural displacements and short-range dependency based on unified joint features projected onto high-dimensional feature space. Specifically, the use of channel attention and Separable TCN, combined with a powerful spatial-temporal attention architecture, results in high-performance accuracy.

A two-stream model incorporating multistage graph convolution with attention and residual connection (GCAR) is proposed to extract spatial-temporal contextual information. Utilizing a channel attention module, the proposed GCAR system enhances attention dynamically, particularly for non-connected skeleton points during specific spatial-temporal events. The channel attention model is applied with each Graph Convolution with Attention (GCA) module to extract global and local features, optimizing model impact. The motion capture process identifies a person's entire body movement during sign language gestures. Pose landmarks from this motion feed into a Graph Convolutional Network (GCN) module, generating a graph-based feature. This feature enters a deep learning module with multiple layers, and a channel attention module selects channel-wise effective features for the GCA module. This process iterates through three GCA modules, with the third module's output concatenated with the residual connection to produce the feature for the first stage of the GCAR module. This series repeats with 22 GCAR modules, resulting in a final feature vector that enhances spatial and temporal contextual information. The enhanced feature vector is fed into the classification module for accurate classification. In contrast, the existing system relies on 2D joint angles-based motion calculation, extracting temporal motion information as a holistic representation of 'body key points' trajectories. Figure 4 depicts the workflow architecture of the proposed method, showcasing the two-stream model. Each stream consists of components, including skeleton or motion, Sep-TCN, GCA, and GCAR. The initial stream generates the pose

stream feature, while the second stream generates the pose motion stream feature. Subsequently, the two stream features are concatenated and fed into the classification module for further processing and classification.

A. POSE ESTIMATION

Skeleton information from the image is effective for hand gesture recognition and human activity recognition because it can hide important human information aimed at protecting security and privacy. The skeleton joint only contains effective information about the hand gesture, including skin colour or texture, aiming to identify a person without the appearance of the hand. The hand texture preserved crucial information about the biometric security system, and the skeleton-based input system can protect this from unauthorised access. In addition, palm prints and fingerprints are considered very sensitive information about a subject, and it is not necessary to explore the skeleton dataset. In the study, face, body, and hand information were incorporated, encompassing more sensitive textures than solely relying on hand gesture data.

A secured hand gesture recognition model was proposed, aiming to conceal the subject's personal information and safeguard their privacy. Various datasets were utilized, extracting skeleton information through diverse methodologies. The dataset consists of 67 skeleton points, including 42 from the left and right hands, 5 from the body, and 20 from the face. Figure 1 illustrates the 67 key points extracted from the entire body, encompassing the body, face, and two hands. Additionally, Table 2 presents the names of the landmarks utilized in the study.

B. MOTION CALCULATION

Dynamic signs were incorporated into various sign language word recognition systems. Preceding the input to the GC model, motion was extracted from the framing sequence to compute effective features for dynamic sign language, emphasizing movement and alignment. This approach aimed to enhance the efficiency of the skeleton-based data structure. In the design and development of the dynamic sign language recognition system, the movement of the joint skeleton was considered a crucial feature directly impacted by motion. A 2-dimensional vector, based on the x and y coordinates from the 67 skeleton key points, was computed by evaluating the joint position difference between consecutive frames. Figure 5 demonstrated the motion calculation visualization, and this can be computed using the following Equations 1:

$$\begin{aligned} Motion_X &= X_t - X_{t+1} \\ Motion_Y &= Y_t - Y_{t+1} \\ Motion_Z &= Z_t - Z_{t+1} \end{aligned} \quad (1)$$

C. SEPARABLE TCN

A distinct variant in the realm of Temporal Convolutional Networks (TCN) is the Separable TCN, also referred to as a depthwise separable TCN. Unlike its traditional

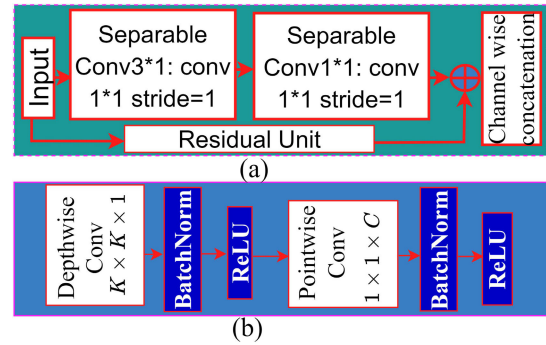


FIGURE 6. (a) SepTCN and (b) Internal structure of the separable TC.

counterpart, the Separable TCN aims to optimize the balance between computational efficiency and model performance by minimizing the number of parameters required. In a conventional Convolutional Neural Network (CNN), the kernel's movement necessitates numerous multiplication operations, leading to escalating computation costs denoted as FLOPS. The Separable TCN strategically decomposes this process into two stages: depth-wise (DW) and pointwise (PW) convolution. The mathematical operations for each time DW requires $K \times K \times 1$, and PW necessitates $1 \times 1 \times C$ filter operations, resulting in a significantly reduced number of operations per kernel during convolutions [42]. This reduction extends to transpose operations in the convolution process, leading to a substantial decrease in the model's total parameters without compromising performance and efficiency. The proposed study applied a 3×1 filter with stride one for the first SepTCN layer and a 5×1 filter with stride 2 for the second Sep-TCN layer, resulting in filter sizes of $3 \times 1 \times 1$ and $1 \times 1 \times 1$ for DW. The Sep-TCN, depicted in Figure 6(a), demonstrates the internal structure of the Sep-TCN, including DW, PW, residual connection, and max-pooling for heightened temporal feature enhancement. It also illustrates the filter size $K \times 1 \times 1$, emphasizing the preservation of the original joint dimension. It focuses on extracting the dominant frame as a feature vector, as showcased in Figure 6(b), which provides detailed insights into the local temporal proximity extraction across a broader scope.

D. GRAPH CONVOLUTION WITH ATTENTION (GCA) MODULE

The motion feature was input into the GCA module to extract local and global range features in terms of spatial and temporal contextual information. Within the GCA module, a Graph convolution layer was employed to enhance long-range dependency, and a set of deep learning layers, including Batch normalization, ReLU, and dropout layer, were used to enhance short-range dependency. Finally, a channel attention layer was applied to enhance the channel-wise effective feature. This module can also deal with the non-connected skeleton point, which leads the system to achieve

high-performance accuracy. Each part of the GCA module is given below.

1) GRAPH CONSTRUCTION

Dynamic datasets for sign language gestures consistently presented a sequence of frames, with each frame's joint skeleton generating a sequence of vectors. In this study, each joint skeleton yielded a 2D vector, forming dynamic sign language gestures comprising varying numbers of frames. A spatial-temporal graph was constructed using joint skeleton information, with the motion of each serving as a graph node and natural connections between joints constituting edges. Figure 1 illustrates an instance of graph construction focusing on the upper body part of the pose. A primary objective is to generate higher-level feature maps on the graph through multiple stages of spatial-temporal graph convolution with attention (GCA) operation [5], [22], [43]. The construction of the graph begins with a fully connected graph comprising N nodes. The edges of the graph are formed based on the weighted adjacency matrix, as per the following equation 2:

$$G = (V, E) \quad (2)$$

Here, V and E denote the vertices and edges of the graph. In the proposed study, there are 67 key points in the full body that are considered vertices in this case, can be expressed as $V = \{v_{(t,i)} \mid t = 1, \dots, T, i = 1, \dots, N\}$ where $N = 67$. In addition, there are 32 frames from each video. Let features for a specific joint skeleton be expressed by $F(v_{ti})$, which is the joint coordinate vectors and estimated confidence of the i th joint on the frame t . The ultimate feature is calculated in two different processes: (i) capture intraframe joint skeleton connection edge that mainly explores the natural connection among the human joint skeleton and (ii) capture the interframe consecutive joint skeleton connection edge. These two processes are explained here as spatial and temporal contextual information. The mentioned edge of the graph is calculated based on the weighted adjacency matrix formula defined in the following Equation 3:

$$A \in \mathbb{R}^{N \times N} \quad (3)$$

Although the edge connection in the same frame can be explored from the natural human-body joint connection, it is difficult to make a fully connected graph from the body and pose information due to its partial connectivity. It means the natural human body is only partially connected as the hand palm surrounding the skeleton and face landmarks are connected. Still, there is no direct connection between the face landmarks and the hand skeleton. Nevertheless, a fully connected graph was constructed to capture joint dependencies through graph learning.

2) GRAPH CONVOLUTION (GC)

Graph Convolution is one type of neural network designed to process the graph structure data. The human body's joint

skeleton carries the graph nature, including the joint skeleton and the natural joint-body connection. The general CNN mainly works with the grid data, such as pixel images, and GC works with nodes and the relationship among nodes explored as edges. It also enables extracting features and patterns from nodes and their local neighbourhoods. The concept of the GC is to aggregate information from neighbouring nodes and use this aggregated information to update the features of the central node. This process is typically performed in multiple layers, allowing the model to capture increasingly complex relationships in the graph.

As previously noted, the proposed system takes a sequence of frames as input, and the sequence number can be expressed at time t . Assume that there are N joint skeletons node V_t in a single frame with a specific time t and skeleton edge in that frame can be written as $Edge_t = \{v_{ti}v_{tj} \mid T = t, (i, j) \in H\}$, where H represent the human body as fully connected graph. As is commonly understood, CNNs typically accept a 2D or 3D grid input and generate a corresponding 2D or 3D grid output. The size of the output shape depends on the stride, filter, etc.; if the stride value is one and uses an appropriate padding size, then the output will be the same size as the input size. The general convolutional neural network equation for a convolutional operator with a $K \times K$ kernel and input data with c channels is given by the following equation 4:

$$F_{(out)}(x) = \sum_{h=1}^K \sum_{w=1}^K data_{in}(S(x, h, w)) \cdot W(h, w) \quad (4)$$

Here, x is the spatial location for a single channel, and the sampling function and weight function are denoted by S and W , respectively. The sampling function mainly enumerates all neighbours of the x location. Here, the weight function generates a weight vector to compute the inner product with the input data in the dimensional real space [22]. The graph convolution (GC) can be defined for the spatial domain with vertices set v_i using Equation 4, as shown in the following equation 5:

$$F_{out}(v_i) = \sum_{(v_j) \in B_i} \frac{1}{Z_{ij}} data_{in}(v_j) \cdot w(l_i(v_j)) \quad (5)$$

Here, Z is the normalization term that is mainly included here to balance the contribution of different nodes. In this case, the degree of the node was utilized to ensure a consistent range of values for each node, where l represents the label map of the corresponding node. The v denotes the set of vertices, and a subset of the node for mapping is denoted B_i . Here, the sampling area denotes the distance between the neighbour vertex. it mainly denoted the distance between neighbour vertices v_i to v_j . The $w(\cdot)$ represents the waited matrix generated function worked as a convolutional layer, which generates the weight vector depending on the user's input terms.

In the proposed study, there is an n -th graph layer in the system and the graph function G_n which mainly takes an input matrix as $data_n \in \mathbb{R}^{N \times F}$, where N is the number of joints,

and F is the feature dimension come from the previous layers. In the starting position, the first layer takes input as $N \times 2T$ body keypoint coordinates, where N is the number of joints in a frame, and T is the total number of frames [17]. The set of trainable weights based on the formula can be written as $W_n \in \mathbb{R}^{F \times \tilde{F}}$.

The GC can be rewrite the 4 as the following Equation 6:

$$data_{n+1} = GC_n(data_n)\sigma(A_n \cdot data_n \cdot W_n) \quad (6)$$

Here, the adjacency matrix, input data and weight matrix are denoted by A , $data$, and W , respectively. The σ is used here as an activation function.

3) CHANNEL ATTENTION

The channel attention model is used here to enhance and highlight the channel-wise features. In order to improve the gen-realization and the discriminative power of the proposed system, the important feature is emphasized, and the less important feature is suppressed. The channel attention employed in the GCA module is illustrated in Figure 4. The GCAR takes the motion feature as input and, after processing the feature with structure data and a deep learning-based layer, aggregates the relationship among the neighbour nodes. The feature matrix generated by GCAR has several channels, and each channel has a corresponding feature map. The main purpose of using channel attention is to refine the GCAR features, aiming to explicitly enhance interdependencies among the channels, yielding dynamic adjustment of the important feature. Channel-wise weight vector allows the system to determine the importance and contribution of each channel. The channel attention generated the weighted feature map using global average pooling for individual channels. The information of each channel is fed into a couple of the fully connected layers to generate the effective feature, then normalized using the batch normalization layer. Following that, a pair of ReLU activation functions were applied to generate positive or zero values. Finally, it will generate the powerful vector by multiplying the output of the activation function with GCAR features [5]. In summary, the channel attention module produces a positive value for the effective feature, and for the less important feature, it produces 1. Multiplication operation changes the GCAR feature by 0, which is not as important as decided by the system. Figure 7 demonstrated the channel attention module where global average pooling in the first place in the module which took N -channel as input. Subsequently, an $N/8$ dense layer was utilized, passing through a batch normalization layer to address issues related to internal covariate shift. In addition, it can also prevent the gradient value from being too small. Next, another fully connected layer with a ReLU activation was applied, aiming to reduce computational complexity. In the final layer, a sigmoid activation function was employed. Figure 7 demonstrated the channel attention module architecture.

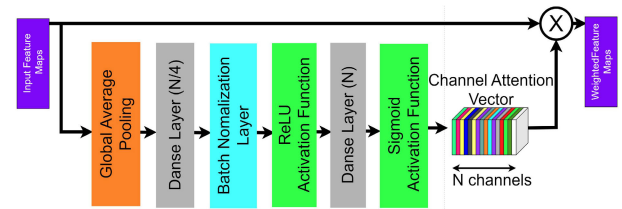


FIGURE 7. Channel attention module.

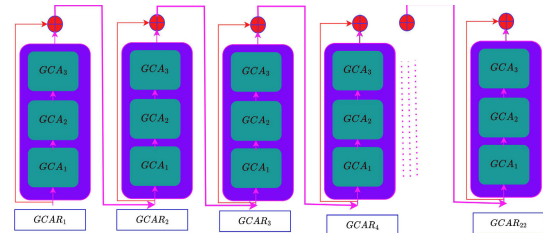


FIGURE 8. Stages of the GCAR model.

E. MULTI-STAGE OF GRAPH CONVOLUTION WITH ATTENTION AND RESIDUAL CONNECTION (GCAR) APPROACH

The residual graph convolution with the attention module comprises stacks of three Graph convolutions with the channel attention (GCA) module as demonstrated in Figure 4. Our GCAR takes the output of the motion feature as an input. The GC performs a convolution on the $N \times T \times C$ initial feature map where vertices' temporal length and the number of channels, respectively. The proposed GCA network is constructed with a graph convolutional layer, ReLU, batch normalization layer, dropout layer and a channel attention module. The GCAR module was constructed of three sequences of the GCA and then concatenated the third GCA output feature with the residual connection. In this study, 22 GCAR modules were employed sequentially to construct the multistage GCAR [17]. Figure 8 illustrates the stages of the network backbone.

F. CLASSIFICATION MODULE

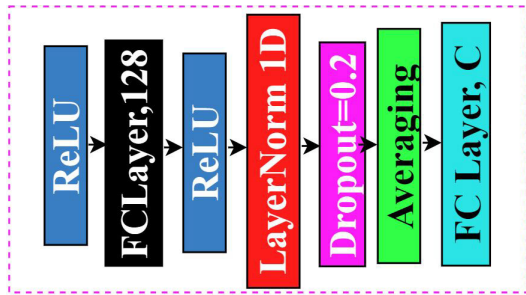
The final feature is fed into the classification module to refine the feature and classification. In the classification module, a composition of coupled ReLU activation, a fully connected layer, data normalization using a batch normalization layer, a dropout layer, and subsequent averaging was employed to convert the matrix into a feature vector. Figure 9 demonstrates the classification module architecture.

V. EVOLUTION AND PERFORMANCE

Various evaluation experiments were conducted on the proposed model using a 4-benchmark dataset to demonstrate its effectiveness, superiority, and power. The experimental setup, evaluation metrics, and performance accuracy were presented along with a state-of-the-art comparison across different datasets.

TABLE 3. Ablation study of the proposed model for the WLASL dataset.

Dataset Class	Top-1 Accuracy			Top-5 Accuracy			Top-10 Accuracy		
	Without Attention (55 Joints)	Without Attention (67 Joints)	Including Attention (67 Joints)	Without Attention (55 Joints)	Without Attention (67 Joints)	Including Attention (67 Joints)	Without Attention (55 Joints)	Without Attention (67 Joints)	Including Attention (67 Joints)
WLASL -100	55.43	56.43	56.50	78.68	79.00	80.00	87.60	87.000	89.14
WLASL-300	38.32	38.00	38.47	67.51	68.75	69.01	79.64	80.00	80.50
WLASL-2000	23.65	23.00	24.10	51.75	52.00	52.62	62.24	62.00	63.32

**FIGURE 9.** Classification module.

A. EXPERIMENTAL SETTING

The samples of a gloss were divided into training and testing sets with a ratio of 4:2, ensuring that each split includes at least one sample per gloss. This splitting strategy aligns with the approach employed in the previous state-of-the-art method. Similarly, for the other three datasets, the dataset splitting technique followed the same strategy, with 70% of the dataset designated as the training dataset and the remaining 30% as the testing dataset. The system was implemented on a machine equipped with a GPU featuring CUDA version 11.7, with NVIDIA driver version 515, and a GeForce RTX 3090 24GB series GPU with 32GB RAM. The learning set utilized a learning rate of 0.003 and a batch size of 32. In the final layer, the number of output neurons was set equal to the number of classes in the dataset, such as WLASL 2000 and ASLLVD 2745. The system underwent 300 epochs with the Adam optimizer on an RTX 3090 machine. Python environment packages included PyTorch [44] for implementing the system with low computational cost for graph convolution and attention, OpenCV, pickle, cv package, pandas package, and other initial processing packages [45], [46].

B. EVALUATION METRIC

The assessment of the proposed models involves calculating the mean scores for top-K classification accuracy, where K is selected from 1, 5, 10, considering all instances of signs. Figure 2 illustrates that distinct meanings often exhibit a close resemblance to sign gestures, leading to potential errors in classification outcomes. Despite this, contextual information can rectify some of these misclassifications. Consequently, opting for top-K predicted labels proves to be a more rational approach for word-level sign language recognition.

C. ABLATION STUDY

The ablation study demonstrated in Table 3 explores the impact of attention mechanisms on the performance of a model across different configurations of the WLASL, including 100 classes, 300 classes and 2000 classes configuration of this dataset. The table mainly consists of the results of three configurations of the WLASL datasets, namely WLASL-100, WLASL-300, and WLASL-1000, each containing varying numbers of American Sign Language (ASL) gestures. Presented here is the ablation study of the proposed model, specifically focusing on the joint motion-based single stream. Three ablation study configurations are considered: “Without Attention (55 Joints),” “Without Attention (67 Joints),” and “Including Attention (67 Joints).” The performance metrics for Top-1, Top-5, and Top-10 accuracy are demonstrated here. A maximum accuracy of 56.50% was achieved for WLASL with 100 class labels, surpassing the performance of the model without attention. An impressive maximum accuracy of 38.47% was achieved for WLASL 300 in terms of Top-5 accuracy. Notably, higher accuracy was obtained with the inclusion of the attention module. The “Without Attention” configurations with 55 and 67 joints were utilized as baseline models, while the “Including Attention” configuration incorporated attention mechanisms. The study reveals that, across all datasets, the inclusion of attention consistently improves accuracy metrics. Notably, the Top-1 accuracy shows substantial gains, indicating the effectiveness of attention mechanisms in capturing critical features for gesture recognition. The results suggest that the attention-enhanced model outperforms its counterparts without attention, showcasing the significance of incorporating attention mechanisms in ASL gesture recognition tasks across different datasets and scales. This comprehensive analysis provides valuable insights for optimizing model architectures in sign language recognition systems.

D. PERFORMANCE ACCURACY AND STATE OF THE ART COMPARISON OF THE WLASL DATASET

Table 4 illustrates the performance of the proposed model, presenting the results for both the single-stream and two-stream models. This comprehensive evaluation of the model’s performance across different datasets and recognition scenarios. The metrics Top-1, Top-3, Top-5, Top-10, and Top-30 signify the accuracy at which the model correctly predicts the top-ranked action class among 100 and 300 classes, respectively. The comparison is made between a Single Stream

model and a more advanced Two Stream model, showcasing the impact of leveraging dual data streams for enhanced recognition. In the WLASL-100 dataset, the Two-stream model consistently outperforms the single-stream counterpart across all evaluation metrics. Notably, at Top-1 accuracy, the Two Stream model achieves a remarkable 63.25% accuracy compared to the Single Stream performance accuracy. This trend continues across higher-ranked predictions, with Top-10 and Top-30 accuracies reaching 90.31% and 97.00%, respectively, for the Two Stream model. For the WLASL-300, WLASL-1000, and WLASL-2000 datasets, a similar pattern emerges, demonstrating the superiority of the Two Stream model. The results highlight the Two Stream model's ability to capture nuanced features and temporal relationships, improving recognition accuracy across varying dataset scales. This comparative analysis is a valuable benchmark for understanding the model's efficacy in different recognition scenarios, guiding further advancements in action recognition research and applications.

Table 5 presents a comprehensive state-of-the-art comparison for the WLASL dataset, offering insights into the performance of various methods across three configurations: WLASL100, WLASL300, and WLASL2000. Evaluation metrics encompass Top-1, Top-5, and Top-10 accuracy percentages, providing a nuanced understanding of model capabilities in predicting correct action classes within specified rank ranges. The performance of ResNet18, detailed in [47], exhibits competitive results across all datasets. Achieving a Top-1 accuracy of 44.54% for WLASL100 and 39.73% for WLASL300, its performance on larger datasets (WLASL300 and WLASL2000) is unavailable (n/a), signaling potential limitations in scalability. In contrast, the Pose-GRU method introduced in [17] records promising results, achieving 46.51%, 76.74%, and 85.86% accuracy for Top-1, Top-3, and Top-5, respectively, on WLASL100. Similarly, for WLASL300, results stand at 33.68%, 64.37%, and 76.05% accuracy for Top-1, Top-3, and Top-5. Outperforming previous methods, Pose-TGCN, as proposed in [17], exhibits superior performance across all datasets. It attains the highest Top-1 accuracy at 55.43% (WLASL100), 38.32% (WLASL300), and 23.65% (WLASL2000). Notably, Pose-TGCN excels not only in single-shot predictions but also in broader classification tasks, which is evident in its high Top-5 and Top-10 accuracies. Additionally, CN-Bert, employed in [26], showcases competitive accuracy, achieving 60.15% (Top-1) and 88.67% (Top-5) on WLASL100, and 42.18% (Top-1) on WLASL300.

In comparison, our proposed method demonstrates compelling results, achieving 63.25%, 88.33%, and 90.31% accuracy for Top-1, Top-3, and Top-5 on WLASL100, and 43.80%, 69.31%, and 80.50% on WLASL300. Furthermore, for WLASL2000, the proposed method records 24.10%, 52.62%, and 63.32% accuracy for Top-1, Top-3, and Top-5.

This comparison elucidates the strengths and weaknesses of each method, emphasizing the proposed model's superior performance. By linking our empirical results with a deeper

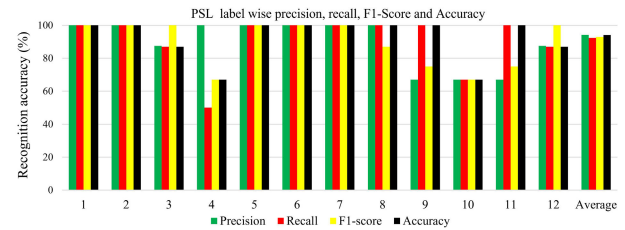


FIGURE 10. Label wise precision, recall and F1-score for the PSL alphabet dataset.

literature review, we provide valuable insights for researchers and practitioners in action recognition, establishing the proposed method as a high-performing solution in the context of the WLASL dataset.

E. PERFORMANCE ACCURACY AND STATE OF THE ART COMPARISON FOR THE PSL DATASET

Table 6 provides a comprehensive performance analysis and state-of-the-art comparison for the PSL dataset, evaluating the proposed Two-Stream model against both its Single Stream counterpart and the dynamic graph-based state-of-the-art model. The proposed Two-Stream model, leveraging Full-body joint points for accurate gesture recognition, achieved an impressive accuracy of 94.01%. To put this result into context, we conducted additional experiments with the Dynamic GCN method proposed by Miah et al. [14], which achieved an accuracy of 90.00%. Furthermore, an experiment was conducted with the Single Stream of our proposed model, resulting in an accuracy of 92.40%. These findings demonstrate the superior performance of our proposed Two-Stream model over both the single-stream and the state-of-the-art Dynamic GCN model. To delve deeper into the label-wise performance, Figure 10 provides a visual representation of the proposed model's accuracy across various labels. Remarkably, the Two-Stream configuration achieved over 85.00% accuracy for most labels, with only three labels exhibiting lower accuracy. This nuanced analysis highlights the model's effectiveness in capturing the complexity of sign language expressions, indicating its potential for diverse real-world applications.

F. PERFORMANCE ACCURACY AND STATE OF THE ART COMPARISON FOR THE MSL

The proposed model also underwent evaluation using the Mexican Sign Language (MSL) dataset, encompassing 3D joint coordinates for the body, face, and hand. In Table 7, we present the performance accuracy of our proposed model in the context of hand gesture recognition, specifically focusing on sign language interpretation using the MSL dataset. A significant benchmark in sign language interpretation was set by the work of the author in [15], where multiple model architectures based on recurrent neural networks, including long short-term memory (LSTM) and recurrent gated units (GRU), were applied. Achieving a commendable

TABLE 4. Performance accuracy with WLASL for various configuration.

Dataset Class	Top-1		Top-3		Top-5		Top-10		Top-30	
	Single Stream	Two Stream	Single Stream	Two Stream	Single Stream	Two Stream	Single Stream	Two Stream	Single Stream	Two Stream
WLASL -100	56.50	63.25	70.00	71.82	80.00	83.33	89.14	90.31	96.12	97.00
WLASL-300	38.47	43.80	56.00	57.00	69.01	69.31	80.00	80.50	88.00	89.00
WLASL-2000	24.10	24.10	44.00	45.00	52.62	52.62	63.32	63.32	78.00	78.50

TABLE 5. State of the art comparison for the WLASL dataset for various configurations.

Method	WLASL100			WLASL300			WLASL2000		
	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10
ResNet18 [47]	44.54	-	-	339.73	-	-	-	-	-
Pose-GRU [17]	46.51	76.74	85.86	33.68	64.37	76.05	-	-	-
PTGCN [17]	55.43	78.68	87.60	38.32	67.51	79.64	23.65	51.75	62.24
CN-BERT [26]	60.15	-	88.67	42.18	-	-	-	-	-
Proposed Model (Two Stream)	63.25	83.33	90.31	43.80	69.31	80.50	24.10	52.62	63.32

TABLE 6. Performance accuracy and comparison for the PSL dataset.

Method Name	Dataset	Joints Key points	Performance Accuracy [%]
Dynamic-GCN [14]	PSL	Full Body	90.00
Proposed Model (Single Stream)	PSL	Full Body	92.40
Proposed Model (Two Stream)	PSL	Full Body	94.1

accuracy of 96.44%, their competence in capturing temporal dependencies within Full Body joint points was evident. In a comparative experiment with the recent state-of-the-art Dynamic-GCN model proposed by Miah et al. [14], the model outperformed traditional RNN architectures, achieving an impressive accuracy of 98.55% on the MSL dataset. This notable result highlighted the effectiveness of dynamic graph-based models in handling the intricacies of sign language recognition. The proposed model, incorporating a Joint Motion Stream module, demonstrated a remarkable improvement in accuracy, reaching an outstanding 99.00%. The Two-Stream configuration surpassed even these results, achieving a remarkable 99.75% accuracy. These findings underscore the effectiveness of the proposed Two-Stream model in capturing intricate sign language gestures, showcasing its potential to significantly enhance communication accessibility for the hearing-impaired community in real-world applications. This discussion establishes the proposed model's superior performance and provides valuable insights into the evolving landscape of sign language recognition models, emphasizing the potential of dynamic graph-based approaches in advancing communication technologies for the hearing-impaired.

G. PERFORMANCE ACCURACY WITH ASLLVD DATASET

Table 8 illustrates the performance accuracy of our proposed model on the large-scale ASLLVD dataset, consisting of approximately 2700 class labels. In contrast to many existing models that are typically evaluated with a limited number of

TABLE 7. Performance accuracy with MSL dataset and state of the art comparison.

Method Name	Dataset Types	Joints Key points	Performance Accuracy [%]
RNN [15]	MSL	Full Body	96.44
Dynamic-GCN [14]	MSL	Full Body	98.55
Proposed Model (Single Stream)	MSL	Full Body	99.00
Proposed Model (Two Stream)	MSL	Full Body	99.75

classes, our method encompasses evaluations for all classes within the dataset. However, achieving high-performance accuracy on this extensive dataset remains a challenge for existing systems. The proposed Two-Stream GCN model exhibits a top accuracy of 34.41%, setting a benchmark for large-scale sign language datasets. To contextualize this result, a comparative analysis was performed with existing models, specifically ST-GCN [23] and 3DGCN [28]. Hammadi et al. introduced a recent method for ASLLVD data, employing a graph-based convolutional neural network with separable 3DGCN layers and a spatial attention mechanism [28]. Their approach achieved 25.05% accuracy for all classes, with notable performance for partial classes but with limited effectiveness for the entire dataset. Comparatively, the proposed Two-Stream GCN model significantly outperformed existing models, achieving 34.41% accuracy. Specifically, ST-GCN [23] generated 16.48%, and the 3DGCN [28] model, which only utilized spatial context information, achieved 25.05%, demonstrating the superiority of our proposed approach. These results emphasize the effectiveness of the Two-Stream GCN model in addressing the challenges posed by large-scale sign language datasets. The notable improvement in accuracy showcases its potential to advance the state of the art in sign language recognition, particularly on datasets with a vast array of class labels. This nuanced analysis not only establishes the performance

TABLE 8. Performance accuracy and state-of-the-art comparison for the ASLLVD dataset.

Method Name	Dataset Types	Features	Class-Label	Performance Accuracy [%]
STGCN [23]	ASLLVD (ASL)	Full Body	2700	16.48
3DGCN [28]	ASLLVD (ASL)	Full Body	2700	25.05
Proposed Model (Single Stream)	ASLLVD (ASL)	Full Body	2700	31.00
Proposed Mode (Two Stream)	ASLLVD (ASL)	Full Body	2700	34.41

superiority of the proposed model but also contributes valuable insights into the evolving landscape of large-scale sign language recognition models.

H. DISCUSSION

In this study, we introduced a sophisticated Two-Stream Graph Convolution with Attention and Residual Connection (GCAR) model, employing a multistage approach to enhance the extraction of spatial-temporal contextual information. The model's architecture includes two stream modules, with the first and second streams receiving joint skeleton and joint motion inputs, respectively. These inputs are then processed through the sep-TCN module before being fed into the GCAR module. An innovative addition to the model involves the integration of a Channel Attention model within each Graph Convolution with Attention (GCA) module, specifically addressing the importance of non-connected skeleton points in certain events. This addition refines spatial-temporal features, emphasizing both global and local aspects crucial for optimal model performance. The GCAR model excels in capturing motion, effectively detecting full-body movements during sign language gestures. To assess the model's performance, we evaluated it on two large-scale datasets and two benchmark datasets. The incorporation of attention mechanisms and residual connections in the GCAR model demonstrated superior accuracy and efficiency, showcasing its state-of-the-art capabilities in comparison to existing models. To manage computational complexity, we utilized Sep-TCN, underscoring the model's proficiency in handling intricate spatial-temporal relationships. Our results indicate that the proposed GCAR model outperforms existing approaches in terms of both accuracy and computational efficiency according to Table 5, Table 8, Table 7, Table 6. The computational complexity of the model across various datasets is detailed in Table 9. We utilized the WLASL [17], ASLLVD [23], MSL [15], and PSL datasets to evaluate the proposed model since these datasets are widely recognized as large-scale datasets for sign language. Numerous researchers have focused on developing systems to enhance the performance accuracy of the WLASL dataset. Notable among them are ResNet18 [47], Pose-GRU [17], PTGCN [17], and CN-BERT [26], considered the latest systems in this field. Du et al. [47] applied vision transfer-based approaches to enhance the

TABLE 9. Computational complexity of the individual dataset for the proposed model.

Dataset Name	BFlops for 1 Batch (32 trial)	Total trial	No Class	Total Batch (Trial/32)	Test Batch	Computation complexity (BFlops)	Parameters (Million)
WLASL	0.13	21089	2000	659	197	25.70	0.69
MSL	0.13	3000	20	93	27	3.51	0.69
PSL	0.13	2700	19	84	25	3.27	0.69
ASLLVD	0.13	9748	2745	304	91	11.58	0.69

performance accuracy of the WLASL dataset, employing pixel-based images and skeleton-based information for sign language words. While they achieved high-performance accuracy with pixel-based information, their results with skeleton information reported a Top-1 accuracy of 44.54% for WLASL100 and 39.73% for WLASL300. Li et al. introduced a pose-based temporal graph neural network in [17], reporting promising results with accuracy figures of 46.51%, 76.74%, and 85.86% for Top-1, Top-3, and Top-5, respectively, on WLASL100. Similarly, for WLASL300, results stand at 33.68%, 64.37%, and 76.05% accuracy for Top-1, Top-3, and Top-5. Pose-TGCN excels not only in single-shot predictions but also in broader classification tasks, achieving the highest Top-1 accuracy at 55.43% (WLASL100), 38.32% (WLASL300), and 23.65% (WLASL2000). Tunga et al. combined GCN with the Bert model in [26], demonstrating competitive accuracy with figures of 60.15% (Top-1) and 88.67% (Top-5) on WLASL100 and 42.18% (Top-1) on WLASL300. In comparison, our proposed method shows compelling results, achieving 63.25%, 88.33%, and 90.31% accuracy for Top-1, Top-3, and Top-5 on WLASL100, and 43.80%, 69.31%, and 80.50% on WLASL300. Furthermore, for WLASL2000, the proposed method records 24.10%, 52.62%, and 63.32% accuracy for Top-1, Top-3, and Top-5 showed in Table 5.

We also acknowledge the significance of comparing our proposed approach with existing state-of-the-art methods, particularly those presented in papers [29] and [30], which also utilize the WLASL dataset for evaluation. Although our approach yields competitive results, it is imperative to address the superior performance of the methods presented in these papers. The method proposed in the paper [30] leverages a hybrid neural network architecture to capture both spatial and temporal features, thereby achieving high accuracy rates for static and dynamic isolated Indian and Russian Sign Language recognition. Similarly, the approach introduced in the paper [29] focuses on multi-semantic discriminative feature learning using a hybrid deep neural architecture, leading to remarkable results in sign gesture recognition tasks. Upon careful analysis, we identify several key differences between our approach and those presented in papers [29], [30]. While our method also emphasizes capturing spatial-temporal contextual information and joint motion through a two-stream multistage graph convolution with attention and residual connection (GCAR) model, we hypothesize that the superior performance of the methods

in papers [29], [30] could be attributed to model architecture and feature extraction technique. The feature representation methods employed in papers [29], [30] might effectively capture discriminative features relevant to sign language recognition tasks, potentially outperforming our approach in certain aspects due to training testing splitting protocols and a number of the class label. While the computational complexity of the mentioned systems was not available, we believe that providing information on the required number of parameters and GFLOPS can aid the reader's understanding. Table 9 summarizes key computational aspects for the evaluated large-scale datasets. The FLOPs for one batch and 32 trials remain constant at 0.13 across all datasets, ensuring a standardized computational baseline. WLASL, with 21,089 total trials and 2,000 classes, showcases a notable computational complexity of 25.70 BFlops for test batches, emphasizing its scale and diversity. By observing the required number of parameters and BFLOPs of the proposed model, we can conclude that our model has high efficiency compared to the existing systems, specifically the methods in [29], [30], for the WLASL dataset. Numerous researchers have focused on developing systems to enhance the performance accuracy of the ASLLVD dataset. Notable among them are STGCN [23] and 3DGCN [28], considered the latest systems in this field. Amorim et al. employed an STGCN for sign language recognition [23] based on the skeleton key points. After evaluating the model with the ASLLVD dataset, they reported 16.48% accuracy. Hammadi et al. introduced a recent method for ASLLVD data, employing a graph-based convolutional neural network with separable 3DGCN layers and a spatial attention mechanism [28] to improve the performance accuracy. Their approach achieved 25.05% accuracy for all classes, with notable performance for partial classes but with limited effectiveness for the entire dataset. Comparatively, the proposed Two-Stream GCN model significantly outperformed existing models, achieving 34.41% accuracy. These results emphasize the effectiveness of the Two-Stream GCN model in addressing the challenges posed by large-scale sign language datasets. The notable improvement in accuracy showcases its potential to advance the state of the art in sign language recognition, particularly on datasets with a vast array of class labels. This nuanced analysis not only establishes the performance superiority of the proposed model but also contributes valuable insights into the evolving landscape of large-scale sign language recognition models. We did not find the computational complexity of the existing system for this dataset, and according to Table 9 ASLLVD, with 9748 total trials and 2745 classes, showcases a notable computation complexity of 11.58 BFlops, emphasizing its scale and diversity. We also showed the state-of-the-art comparison for the MSL dataset 7. Perez et al. employed a hybrid model including LsTM, RNN and GRU approach in [15], and they reported 96.44% accuracy; their competence in capturing temporal dependencies within Full Body joint points was evident. In a comparative experiment with the

recent state-of-the-art Dynamic-GCN model proposed by Miah et al. [14], the model outperformed traditional RNN architectures, achieving an impressive accuracy of 98.55% on the MSL dataset. This notable result highlighted the effectiveness of dynamic graph-based models in handling the intricacies of sign language recognition. The proposed model, incorporating a Joint Motion Stream module, demonstrated a remarkable improvement in accuracy, reaching an outstanding 99.00%. The Two-Stream configuration surpassed even these results, achieving a remarkable 99.75% accuracy. We also did not find the computational complexity, and we reported computational complexity in terms of the required parameters and BFLOPS. MSL, featuring 3000 trials and 20 classes, exhibits a comparatively lower complexity of 3.51 BFlops, indicating a more focused dataset where PSL needs 3.27 BFLOPS.

VI. CONCLUSION

We proposed a Graph Convolution with Attention and Residual Connection (GCAR) model featuring a two-stream multistage architecture designed for the extraction of spatial-temporal contextual information. The first stream utilized joint skeleton points as input, incorporating sep-TCN to stabilize computational complexity. Through the sequential application of GCN, a deep learning module, and a channel attention module, the first stream generated the final feature. Simultaneously, the second stream processed consecutive frame joint motion information, following a similar procedure to produce its final feature, which was then fused with the first stream's output. The proposed approach effectively captured motion, enabling the detection of full-body movements during sign language gestures. The utilization of sep-TCN aimed at stabilizing computational complexity, while the incorporation of channel attention sought to enhance the significance of non-connected skeleton points in specific spatial-temporal events. Each GCN module incorporated the channel attention model to extract both global and local features, contributing to model optimization. The multistage integration of GCAR further augmented spatial and temporal contextual information, resulting in the generation of an effective feature vector through the fusion of graph and general CNN techniques. The proposed model achieves the goal by achieving high-performance accuracy across various large-scale sign language datasets, which proves the superiority of the proposed system. Notably, The proposed model addressed the challenge of inter-gesture similarity and insufficient skeleton information encountered in datasets focused solely on hand skeletons, as the proposed approach considered the entire body, face, and both hands, encompassing 67 key points. In the future, the deployment of this system as a practical sign language recognizer is envisioned. Further enhancements to performance accuracy will be pursued through the incorporation of multiple camera-based recognition techniques. This evolution of the proposed method aligns with the goal of creating a robust

and versatile solution for real-life sign language recognition applications.

ABBREVIATIONS

GCN	Graph Convolutional Network
GCAR	Graph convolution with attention and residual connection
SLR	Sign Language Recognition
ASL	American Sign Language
BSL	Britain sign Language
SLR	Sign Language Recognition
ASLLVD	American Sign Large Video Dataset
MSL	Mexican sign language
RNN	Recurrent Neural Network
LSTM	Long short term memory
TGCN	Temporal graph convolutional network
ML	Machine learning
DL	Deep learning
JFM	Jacobi-fourier moments
CTC	Connectionist temporal classification
3D-ResNet	3D convolutional residual network
soft-DTW	Soft dynamic time warping
BFlops	Billion floating-point operations
WER	worderror rate
LS-HAN	Hierarchical attention network with latent space
Img	Image
SMTRM	Shape analysis of motion trajectories on riemannian manifold
Res-TCN	Residual Connection with temporal convolutional neural network

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